Enrolment number: _____

# MID TERM EXAMINATION, 2026

**Branch & Academic Year: CSE** | **Semester: IV**

**Subject: COMPUTER NETWORKS** | **Subject Code: CS206C** | **Instructor Code: SPK**

**Max Marks: 30 Marks** | **Duration: 60 Minutes** | **Date: 26-02-2025**

***Instruction:*** ***First of all write enrolment number on top of question paper.***

**Part A (Solve any 10)** **(Each question Carries 2 marks)** **20 marks**

1. Why is HTTP said to be an application-layer protocol even though it uses TCP?
2. What problem does DNS solve and why is it implemented at the application layer?
3. Why does e-mail use multiple application-layer protocols instead of one?
4. Explain how Web caching reduces user-perceived delay.
5. Why are cookies required if HTTP is stateless?
6. Why are CDNs used in video streaming applications?
7. Why is P2P file distribution more scalable than client-server distribution?
8. Why is TLS considered an application-layer enhancement rather than a transport protocol?
9. Distinguish between bandwidth-sensitive and elastic applications.
10. Why are port numbers required in addition to IP addresses?
11. HFC, DSL, and FTTH are all used for residential access. For each of these access technologies, provide a range of transmission rates and comment on whether the transmission rate is shared or dedicated.Mention a disadvantage with legged locomotion.
12. What is self-replicating malware?

**Part B(Solve any 4)** **(Each question Carries 2.5 marks)** **10 marks**

13. Describe the client–server and peer-to-peer application architectures. Compare their advantages, limitations, and use cases..
14. Explain the role of the transport layer services for applications. Compare TCP and UDP in terms of reliability, timing, throughput, and security.
15. Explain DNS architecture and its importance in Internet operation. Why is DNS designed as a distributed hierarchical system?
16. Explain Web caching and Content Distribution Networks (CDNs). How do they improve performance and scalability of Internet applications?
17. Explain how HTTP works. Describe how a Web page is retrieved and discuss why HTTP is stateless.